

ROOT Introductory Exercises

Download and install ROOT on your operating system.

<https://root.cern.ch/downloading-root>

Once finished, find the ROOT primer

https://root.cern/get_started/

or the intro student course.

<https://github.com/root-project/student-course>

In the primer, work your way through ROOT basics, ROOT macros, Graphs, and Histograms. Then, attempt the exercises below.

- 1.) Create a histogram of random values between 0 and 1000 that correspond to a Gaussian curve.

Create a fit of this Gaussian and output the parameters.

- 2.) Write either a macro or a compiled ROOT code that plots the steady state value of p as a function of r , where p is the recursive function below. Have a starting value $p = 0.02$.

$$p_{n+1} = rp_n(1-p_n)$$

Start by writing a code that calculates p for n repetitions. (Output to screen is ok.)

Use a 2-dimensional histogram to see how the bound value behaves as you increase r between 1 and 4. The increments should be at least 0.1 units.

- 3.) Write a program/macro that completes the following steps:
 - a. Define the three corners of an equilateral triangle. Let's call these Points 1, 2, and 3.
 - b. Randomly select a point in the interior of the triangle. Let's call this Point A.
 - c. Randomly select Point 1, 2, or 3, and draw a point halfway between Point A and the selected point. Let's call this Point B.
 - d. Randomly select Point 1, 2, or 3, and draw a point halfway between Point B and the selected point.
 - e. Continue the process of drawing points halfway between your newest point and one of the three corners. Draw at least 10,000 points. You

may find that you need to shrink the marker size in your plot to get the best quality image. Enjoy!